Sure. Here is the **complete syllabus/roadmap** we will cover for your Geospatial Application topic, including **QGIS, PyQGIS, GIS, Remote Sensing, CNN, and the project**.

Complete Roadmap

PART 1 — Geospatial Fundamentals

1. Introduction to Geospatial Technology

- What is geospatial technology?
- What is geospatial data?
- Spatial vs non-spatial data
- Location information
- Latitude and longitude
- Geographic coordinates

2. Types of Geospatial Data

- Vector data
  - Point
  - Line
  - Polygon
- Raster data
- Raster vs vector
- Attributes
- Spatial features

3. Coordinate Reference Systems

- CRS
- Geographic Coordinate System
- Projected Coordinate System
- WGS84
- EPSG
- Why CRS is important
- Reprojection

PART 2 — GIS

4. GIS Basics

- What is GIS?
- Components of GIS
- GIS workflow

- GIS applications
- Layers
- Attribute tables
- Spatial relationships

5. Spatial Analysis

- Querying spatial data
- Selection
- Distance analysis
- Overlay analysis
- Buffer
- Clip
- Intersection
- Union
- Dissolve
- Spatial join

PART 3 — QGIS ⭐

6. Introduction to QGIS

- What is QGIS?
- QGIS vs GIS
- QGIS interface
- Projects
- Browser panel
- Layers panel
- Map canvas

7. Working with Layers

- Add vector layer
- Add raster layer
- Shapefiles
- GeoJSON
- CSV with coordinates
- Layer properties
- Attribute tables

8. QGIS Symbology

- Single symbol

- Categorized symbology
- Graduated symbology
- Color/size based on attributes
- Labels
- Map styling

9. QGIS Vector Operations

- Select features
- Buffer
- Clip
- Intersection
- Union
- Dissolve
- Merge
- Spatial join

10. QGIS Raster Operations

- Raster layers
- Raster calculator
- Raster statistics
- Reclassification
- Raster clipping
- Raster overlay
- DEM analysis

11. QGIS Map Creation

- Map composition
- Scale
- Legend
- North arrow
- Labels
- Coordinate grid
- Exporting maps

PART 4 — PYTHON GIS / PyGIS ⭐

12. Python for Geospatial Data

- Why Python for GIS?
- Reading geographic data

- Processing geographic data
- Automating GIS tasks

13. Important Python Libraries

- NumPy
- Pandas
- GeoPandas
- Shapely
- Rasterio
- PyProj
- Fiona
- Folium
- Matplotlib

14. GeoPandas

- GeoDataFrame
- Reading shapefiles
- Reading GeoJSON
- Attribute manipulation
- Filtering
- Spatial operations
- CRS handling
- Exporting data

15. Shapely

- Point
- LineString
- Polygon
- Geometry operations
- Distance
- Intersection
- Buffer

16. Rasterio

- Reading raster data
- Raster properties
- Bands
- Pixel values
- Writing raster data
- Raster processing

PART 5 — PyQGIS ⭐

17. PyQGIS Basics

- What is PyQGIS?
- QGIS Python Console
- QGIS Python API
- Python + QGIS

18. Working with QGIS through Python

- Load layers
- Access features
- Access attributes
- Create layers
- Modify layers
- Modify geometries
- Save/export layers

19. QGIS Processing with Python

- Run QGIS algorithms using Python
- Buffer
- Clip
- Dissolve
- Intersection
- Raster operations
- Automating workflows

20. Why PyQGIS?

Understand the difference:

```
QGIS
 ↓
Manual GIS operations

PyQGIS
 ↓
Automated GIS operations
```

PART 6 — Remote Sensing ⭐

21. Remote Sensing Basics

- What is remote sensing?
- How satellites observe Earth
- Sensors
- Platforms
- Active vs passive sensing

22. Satellite Imagery

- Satellite images
- Image bands
- Pixels
- Image resolution

23. Types of Resolution

- Spatial resolution
- Spectral resolution
- Temporal resolution
- Radiometric resolution

24. Types of Satellite Images

- RGB
- Multispectral
- Hyperspectral
- Panchromatic

25. Major Satellite Data

- Landsat
- Sentinel
- Google Earth Engine
- Google Earth
- Satellite image datasets

PART 7 — Image Processing

26. Digital Images

- Pixels
- Channels
- Image dimensions
- RGB representation

- $256 \times 256 \times 3$

27. Image Preprocessing

- Resizing
- Normalization
- Cropping
- Image patches
- Noise
- Data augmentation

28. Remote-Sensing Indices

- NDVI
- NDWI
- Other useful spectral indices
- Interpretation of index values

PART 8 — CNN ⭐⭐⭐

29. Neural Network Basics

- Neuron
- Weights
- Bias
- Activation function
- Forward propagation
- Loss

30. CNN Introduction

- Why CNN?
- CNN vs traditional neural network
- Feature extraction

31. Convolution

- Kernel/filter
- Feature map
- How convolution works
- 3×3 filter
- Multiple filters

32. CNN Numericals ⭐

Formula:

$$\boxed{O = \left\lfloor \frac{N - F + 2P}{S} \right\rfloor + 1}$$

We'll cover:

- Input size
- Filter size
- Padding
- Stride
- Output size
- Output channels
- Multiple convolution layers

33. Padding

- Valid padding
- Same padding
- Zero padding

34. Stride

- Stride 1
- Stride 2
- Effect on output

35. Activation Functions

- ReLU
- Sigmoid
- Softmax

36. Pooling

- Max pooling
- Average pooling
- Pooling numericals
- Why pooling?

37. Complete CNN Architecture

```
Input
 ↓
Convolution
 ↓
ReLU
```

```
 ↓
Pooling
 ↓
Convolution
 ↓
ReLU
 ↓
Pooling
 ↓
Flatten
 ↓
Fully Connected
 ↓
Softmax
 ↓
Prediction
```

We'll calculate dimensions at every stage.

PART 9 — CNN Training

38. Dataset Preparation

- Dataset
- Labels
- Training set
- Validation set
- Test set

39. Training

- Forward propagation
- Loss
- Backpropagation
- Gradient descent
- Epoch
- Batch
- Learning rate

40. Classification Loss

- Binary cross-entropy
- Categorical cross-entropy
- Softmax + cross-entropy

41. Model Evaluation

- Accuracy

- Precision
- Recall
- F1-score
- Confusion matrix

PART 10 — Geospatial ML

42. Land-Cover Classification ⭐

Classes such as:

- Vegetation
- Water
- Buildings
- Roads
- Agriculture
- Bare land

43. Satellite Image Classification

```
Satellite Image
       ↓
Preprocessing
       ↓
CNN
       ↓
Feature Extraction
       ↓
Classification
```

44. Land-Cover Mapping

- Classification maps
- Raster maps
- Geographic visualization
- Accuracy assessment

PART 11 — Vegetation Analysis 🌳

45. Vegetation Detection

- Identifying vegetation
- RGB vegetation detection

- NIR
- Spectral information

46. NDVI ⭐

- NDVI concept
- NDVI formula
- NDVI calculation
- NDVI interpretation
- Vegetation maps

47. Vegetation Percentage

- Calculate vegetation coverage
- Compare different areas
- Create vegetation maps

PART 12 — Change Detection ⭐

48. Temporal Analysis

- Comparing images from different years
- Multi-temporal satellite data

49. Change Detection

```
Image 2020
     ↓
   Compare
     ↑
Image 2026
     ↓
Change Map
```

Applications:

- Deforestation
- Urbanization
- Vegetation loss
- Agricultural changes
- Water-body changes

PART 13 — GIS + CNN Integration ⭐⭐⭐

50. Combining CNN and GIS

Remember:

> **CNN tells WHAT.**

> **GIS tells WHERE.**

> **Python processes/automates.**

> **QGIS visualizes and performs GIS analysis.**

Complete flow:

```
Satellite Image
       ↓
      CNN
       ↓
Land-Cover Classification
       ↓
Geospatial Coordinates
       ↓
      GIS
       ↓
      QGIS
       ↓
Spatial Analysis
       ↓
Final Map
```

PART 14 — Geospatial Recommendation System ⭐

51. Recommendation System

Example:

> **Where should trees be planted?**

52. Factors

- Existing vegetation
- Soil
- Rainfall
- Temperature
- Water availability

- Elevation
- Slope
- Land use
- Distance from roads/water
- Protected areas

53. Suitability Analysis

```
Soil Layer
Rainfall Layer
Water Layer
Slope Layer
Vegetation Layer
       ↓
   GIS Overlay
       ↓
Suitability Score
       ↓
Rank Locations
       ↓
Recommendation
```

PART 15 — Image Segmentation ⭐

54. Classification vs Segmentation

Classification:

> "This image contains vegetation."

Segmentation:

> "These exact pixels are vegetation."

55. Semantic Segmentation

- Pixel-level classification
- Land-cover segmentation

56. U-Net

- U-Net architecture
- Encoder
- Decoder
- Skip connections

- Why U-Net is useful for satellite images

PART 16 — Advanced CNN

57. Transfer Learning

- Pretrained models
- Feature reuse
- Fine-tuning

58. CNN Architectures

- ResNet
- EfficientNet
- MobileNet

59. Model Improvement

- Data augmentation
- Hyperparameter tuning
- Regularization
- Batch normalization
- Dropout

PART 17 — Actual Project

60. Project Problem Statement

61. Project Objectives

62. Project Architecture

63. Dataset Selection

64. Data Collection

65. Data Preprocessing

66. CNN Model

67. GIS Integration

68. QGIS Visualization

69. PyQGIS Automation

70. Results

71. Evaluation

72. Final Application

PART 18 — Project Report + Viva

73. Report

- Abstract
- Introduction
- Problem statement
- Objectives
- Literature review
- Methodology
- Architecture
- Dataset
- Implementation
- Results
- Limitations
- Future scope
- Conclusion

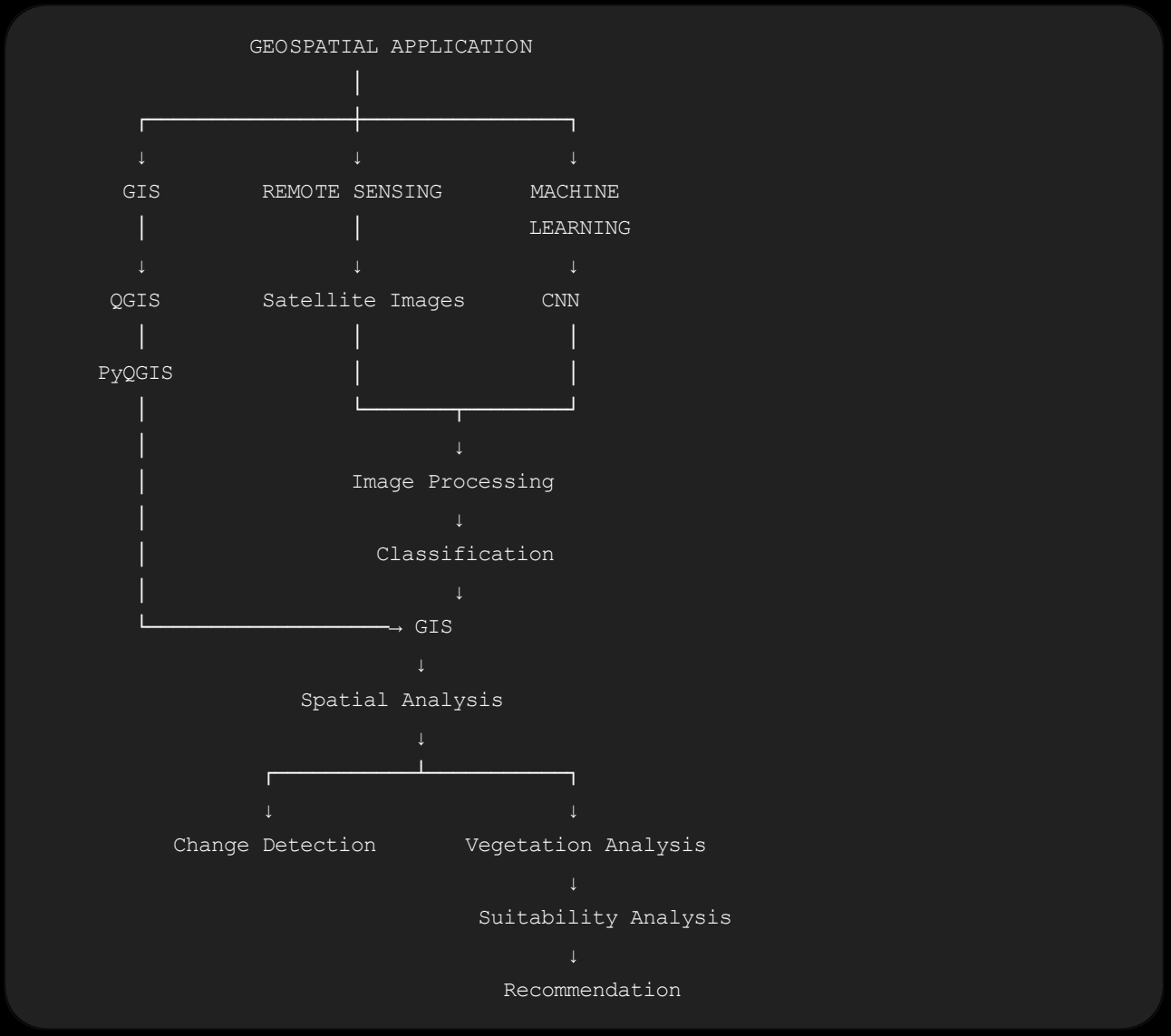
74. Viva

We'll prepare answers for:

- Why GIS?
- Why QGIS?
- Why PyQGIS?
- Why remote sensing?
- Why satellite imagery?
- Why CNN?
- Why $256 \times 256 \times 3$?
- Why 3×3 filter?
- Padding?
- Stride?
- NDVI?
- Land-cover classification?

- Change detection?
- CNN vs U-Net?
- How does GIS integrate with CNN?
- How do you recommend tree-planting locations?
- What are the limitations?

🔥 The complete big picture



So **nothing is missing now**: **GIS + QGIS + PyQGIS + Python GIS + Remote Sensing + Satellite Data + CNN + NDVI + Land Cover + Change Detection + Recommendation System + U-Net + final project** are all included.